

A
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On
“Youtube Analytics Dashboard”

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In

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Certificate

This is to certify that the Project Report entitled “Youtube Analytics Dashboard” submitted by Shagun Gupta & Reansh Singh, students of United College of Engineering and Research, Naini, Prayagraj, is an original work carried out at Summer Training 2026, UGI, in partial fulfillment of academic requirements. The training and project were completed under the supervision and guidance of Mr. Ajai Maurya. The matter embodied in this report is original and has not been submitted to any University/Institute for the award of any degree

Date: July 03,2026

Supervisor:
Mr. Ajai Maurya

Declaration

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Abstract

In the era of digital media, YouTube has emerged as one of the world's largest content-sharing platforms, generating enormous volumes of data related to subscribers, video views, content categories, audience demographics, and revenue. Analysing such large datasets manually is both challenging and time-consuming. Business Intelligence (BI) tools provide an effective solution by transforming raw data into meaningful insights through interactive visualizations and analytical dashboards.

The objective of this Summer Training Project is to design and develop a **YouTube Analytics Dashboard** using **Microsoft Power BI** to analyse global YouTube channel statistics and present the information in an intuitive, interactive, and visually appealing manner. The project focuses on converting raw and unstructured data into actionable business insights that assist users in understanding subscriber distribution, category-wise performance, country-wise channel analysis, historical growth trends, and financial earnings. The dataset used for this project, "**Cleaned_Youtube_Stats**," was initially examined and prepared using **Microsoft Excel**, where data validation and preliminary formatting were performed. The cleaned dataset was then imported into **Microsoft Power BI**, where data transformation, modeling, and visualization techniques were applied. Various interactive visualizations, including **Donut Charts, Bar Charts, Line Charts, Area Charts, Waterfall Charts, Funnel Charts, Tables, and Decomposition Trees**, were implemented to provide comprehensive analytical insights.

Interactive filtering and cross-analysis capabilities were also incorporated to enhance user experience and support efficient data exploration.

The developed dashboard successfully highlights key performance indicators such as **subscriber demographics, geographical distribution of YouTube channels, content category analysis, and yearly earnings**. It enables users to identify market trends, compare channel performance across different countries and categories, and gain valuable insights for data-driven decision-making.

This project demonstrates the practical application of **Business Intelligence, Data Visualization, and Data Analytics** techniques using Microsoft Power BI and Microsoft Excel. It also enhances understanding of data preprocessing, dashboard design, interactive reporting, and analytical thinking, making it a valuable learning experience in the field of Business Intelligence and Data Analytics.

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Objectives

1. **Develop an interactive YouTube Analytics Dashboard** using Microsoft Power BI to transform raw YouTube statistics into meaningful, visually appealing, and data-driven insights for effective analysis.
2. **Analyse global YouTube channel performance** by examining key metrics such as subscribers, video views, channel rankings, content categories, uploads, and financial earnings to identify performance trends and patterns.
3. **Design an intuitive and user-friendly dashboard interface** that enables users to explore complex datasets through interactive visualizations, filters, and multiple report pages for enhanced decision-making.
4. **Perform data preprocessing and transformation** by utilizing Microsoft Excel and Power BI to clean, validate, standardize, and organize the dataset, ensuring data accuracy, consistency, and reliability for analysis.
5. **Implement a variety of Power BI visualizations**, including Donut Charts, Bar Charts, Line Charts, Area Charts, Funnel Charts, Waterfall Charts, Tables, and Decomposition Trees, to represent different aspects of YouTube analytics effectively.
6. **Enable dynamic data exploration** by incorporating interactive slicers and cross-filtering features that allow users to analyse subscriber distribution, country-wise performance, category-wise statistics, and financial metrics from multiple perspectives.
7. **Evaluate historical trends and business performance** by analysing channel creation years, subscriber growth patterns, geographical distribution, and yearly earnings to uncover meaningful insights and support data-driven decision-making.

8. **Demonstrate the practical application of Business Intelligence (BI) and Data Visualization techniques** using Microsoft Power BI, showcasing how analytical dashboards can simplify complex datasets and improve information accessibility.
9. **Enhance practical knowledge of data analytics and business intelligence concepts** by applying data cleaning, transformation, visualization, dashboard design.

Tools and Technologies

S. No.	Tool / Technology	Purpose / Description
1	Microsoft Excel	Used for initial data inspection, validation, preprocessing, formatting, and preparation of the raw YouTube dataset before importing it into Power BI.
2	Microsoft Power BI Desktop	Served as the primary Business Intelligence tool for data modelling, dashboard creation, interactive reporting, and visualization of YouTube analytics.
3	Power Query Editor	Performed ETL (Extract, Transform & Load) operations, including cleaning, transforming, organizing, and preparing the dataset for analysis.
4	Data Analysis Expressions (DAX)	Used to create calculated measures and dynamic aggregations for subscriber counts, yearly earnings, and other analytical metrics.
5	Data Visualization Components	Utilized interactive visualizations such as Donut Charts, Bar Charts, Line Charts, Area Charts, Funnel Charts, Waterfall Charts, Tables, and Decomposition Trees to present analytical insights.
6	Interactive Slicers & Filters	Enabled dynamic filtering based on Country and Category, allowing users to explore and analyse data interactively across multiple report pages.
7	Cleaned_YouTube_Stats Dataset	The primary dataset containing YouTube channel statistics, including subscribers, video views, earnings, uploads, categories, countries, and channel information.
8	Microsoft Windows	Provided the development environment for executing Microsoft Excel and Power BI Desktop, supporting the complete dashboard development process.

Dashboard Functionality / Modules

The **YouTube Global Analytics Dashboard** is designed as a multi-page interactive Business Intelligence solution that enables users to analyze global YouTube channel statistics through dynamic visualizations and intuitive reporting. The dashboard is organized into different modules, each focusing on a specific aspect of data analysis. These modules collectively provide meaningful insights into subscriber demographics, geographical distribution, channel performance and financial earnings

The following are the primary modules of the dashboard:

1.1 Subscriber Analytics Module

This module focuses on analysing subscriber-related statistics across different YouTube channels. It provides interactive visualizations that help users understand subscriber distribution based on content categories, geographical regions, and channel creation years.

The module includes visualizations such as Donut Charts, Bar Charts, Line Charts, and Data Tables to present subscriber insights effectively. Interactive filtering allows users to explore subscriber statistics based on selected countries and content categories.

1.2 Revenue Analytics Module

The Revenue Analytics Module provides a comprehensive analysis of the financial performance of YouTube channels. It focuses on yearly earnings, category-wise revenue distribution, and country-level financial contributions.

Advanced Power BI visualizations, including Funnel Charts, Waterfall Charts, Area Charts, and Decomposition Trees, are used to analyse earnings from multiple perspectives. This module enables users to identify high-performing content categories and countries contributing significantly to overall revenue.

1.3 Interactive Dashboard Features

The dashboard incorporates several interactive features that enhance user experience and facilitate efficient data exploration. Users can dynamically filter reports, compare multiple analytical dimensions, and interact with visualizations to gain deeper insights.

Key Features

The YouTube Global Analytics Dashboard has been developed to provide an interactive, user-friendly, and data-driven platform for analyzing global YouTube channel statistics. The dashboard incorporates several advanced Business Intelligence features that enable users to explore, compare, and interpret complex datasets efficiently. The key features of the dashboard are described below.

2.1 Interactive Dashboard

The dashboard provides an intuitive and interactive interface that enables users to explore YouTube statistics through multiple report pages. The interactive design allows users to navigate seamlessly between different analytical sections while maintaining a consistent and user-friendly experience.

2.2 Dynamic Filtering

Interactive slicers have been implemented to allow users to filter the dashboard based on Country and Category. These filters dynamically update all related visualizations, enabling customized analysis according to user requirements.

2.3 Comprehensive Data Visualization

The dashboard utilizes a wide range of Power BI visualizations to represent analytical information in a meaningful and visually appealing manner. Different chart types have visualizations used:

Donut chart, Funnel chart, Waterfall chart, Matrix chart, Decomposition tree, Clustered Bar chart, Area chart, Line chart.

2.4 Subscriber Analytics

The dashboard provides detailed insights into subscriber distribution across various content categories and countries. It also analyses subscriber growth trends based on channel creation years, enabling users to identify high-performing categories and geographical regions

Performance and Evaluation

The developed YouTube Global Analytics Dashboard was evaluated based on its functionality, responsiveness, visualization quality, and ability to generate meaningful business insights.

The dashboard successfully transformed raw YouTube statistics into an interactive analytical platform, enabling efficient exploration of subscriber demographics, geographical distribution, and financial performance.

The dashboard also demonstrated efficient performance while processing the available dataset. Navigation between report pages was smooth, and all charts updated instantly when slicers were applied, providing a seamless user experience.

The use of Power BI's interactive filtering and cross-highlighting features ensured that all visualizations responded dynamically to user selections, allowing users to analyse data from multiple perspectives. The implemented visualizations accurately represented the dataset and improved the readability of complex information.

Overall, the dashboard achieved the intended project objectives by providing an efficient and interactive Business Intelligence solution for analysing global YouTube statistics. It effectively supports data-driven decision-making through meaningful visualizations and intuitive reporting.

Limitations

Although the YouTube Global Analytics Dashboard provides valuable insights through interactive visualizations and Business Intelligence techniques, it has certain limitations:

1. The dashboard is based on a static dataset and does not reflect real-time changes in YouTube statistics.
2. The project does not integrate with the YouTube Data API, limiting automatic data updates and live analysis.
3. The accuracy of the analysis depends entirely on the quality and completeness of the available dataset.
4. The dashboard focuses on descriptive analytics and does not include predictive models or machine learning-based forecasting.
5. The analysis is limited to the selected attributes available in the dataset, such as subscribers, earnings, categories, and countries.

Despite these limitations, the dashboard effectively demonstrates the practical application of Microsoft Power BI for interactive data visualization and Business Intelligence, providing meaningful insights into global YouTube channel performance.

Future Work

The YouTube Global Analytics Dashboard successfully demonstrates the use of Business Intelligence techniques for analysing YouTube channel statistics through interactive visualizations and meaningful insights. However, there is significant scope for further enhancement to improve the functionality, analytical capabilities, and overall user experience of the dashboard.

One of the major improvements would be the integration of the YouTube Data API. Instead of relying on a static dataset, the dashboard could retrieve live data directly from YouTube, allowing users to analyse real-time statistics such as subscriber count, video views, engagement metrics, and earnings. This would ensure that the dashboard always reflects the most recent information.

The dashboard can also be extended by including additional Key Performance Indicators (KPIs) such as average views per video, engagement rate, subscriber growth rate, likes-to-views ratio, comments analysis, and audience retention metrics. These indicators would provide a more comprehensive understanding of channel performance and audience behavior.

The dashboard could further be enhanced by integrating advanced Artificial Intelligence (AI) features available in Power BI, such as AI-powered visualizations, anomaly detection, smart narratives, and automated insights. These features would help users identify hidden patterns, unusual trends, and significant changes within the dataset more efficiently.

In conclusion, the implementation of these enhancements would transform the current dashboard into a more comprehensive, intelligent, and real-time Business Intelligence solution capable of supporting advanced analytics and strategic decision-making for YouTube content analysis.

References

The successful completion of this project was supported by various authentic and reliable sources, including official documentation, technical guides, research books, and online learning resources. These references provided valuable knowledge on Microsoft Power BI, Microsoft Excel, Business Intelligence, Data Visualization, Data Analysis Expressions (DAX), and dashboard development. The dataset and supporting materials used throughout the project were also obtained from credible sources. The following references were consulted during the design, implementation, and documentation of the YouTube Global Analytics Dashboard project.

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